# Week 4 Module

## Question 01

Question - 1

- Take input (N)
- Use for loop (0 - N)
- Print num followed by space
- Stop at the end.

```java
import java.util.Scanner;

public class print_num {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        for (int i = 1; i <= n; i++) {
            System.out.print(i + " ");
        }
    }
}
```

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL

Windows PowerShell
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PS C:\Users\Neeraj\Desktop\college\code> cd "c:\Users\Neeraj\Desktop
}
5
1 2 3 4 5 PS C:\Users\Neeraj\Desktop\college\code>
```

## Question 02

Question - 2

- Take input (N)
- make variable [Sum = 0;]
- use for loop [0-N]
- Add each num to sum
- print Sum.

```java
import java.util.Scanner;

public class sum_of {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        long sum = 0;

        for (int i = 1; i <= n; i++) {
            sum += i;
        }

        System.out.println(sum);
    }
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```
                         ^
tempCodeRunnerFile.java:2: error: class, interface, enum, or record expected
        for (int i = 1; i <= n; i++) {
                                ^
3 errors
PS C:\Users\Neeraj\Desktop\college\code> cd "c:\Users\Neeraj\Desktop\college\code\
5
15
PS C:\Users\Neeraj\Desktop\college\code>
```

**Question 03**

Question - 3

- Take input A, B
- make variable [result = 1;]
- use for loop B time.
- multiply result by A
- print result.

```java
import java.util.Scanner;

public class power {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int a = sc.nextInt();
        int b = sc.nextInt();

        long result = 1;

        for (int i = 1; i <= b; i++) {
            result *= a;
        }

        System.out.println(result);
    }
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```
3 errors
PS C:\Users\Neeraj\Desktop\college\code> cd "c:\Users\Neeraj\Desktop\co
5
15   Focus folder in explorer (ctrl + click)
PS C:\Users\Neeraj\Desktop\college\code> cd "c:\Users\Neeraj\Desktop\co
2 5
32
PS C:\Users\Neeraj\Desktop\college\code>
```

# Complexity Analysis :-

| Question | Time Complexity | Space Complexity |
|---|---|---|
| Print 1 to N | O(N) | O(1) |
| Sum of N Numbers | O(N) | O(1) |
| $A^B$ Using Loop | O(B) | O(1) |